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EXTENDED-ABSTRACT

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Abstract

We present the Ommie Deployable System (DS), a replicable, autonomous platform for long-term, in-the-wild mental health applications with the Ommie robot. Ommie DS builds on prior anxiety-focused deployments by introducing robust hardware, enhanced sensing, modular software, a companion tablet, and wireless multi-device architecture to support daily deep-breathing interactions in homes. Designed using off-the-shelf components and rapid-prototyped enclosures, the system enables reliable multi-week use, remote monitoring, and easy customization. By providing a durable, open, and researcher-friendly platform, Ommie DS supports scalable, real-world study of HRI for mental health and well-being.

CCS Concepts

• Computer systems organization → Robotics; • Human-centered computing → Interaction devices.

Keywords

social robots, mental health, in-the-wild, HRI, robot systems

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1 Introduction

With the rise of robots in homes, workplaces, medical centers, and more, the demand for longitudinal, in-the-wild human-robot interaction (HRI) research is increasing [10]. Unlike short-term laboratory studies, in-the-wild research exposes systems to the human needs, daily challenges, and user preferences that shape how robots must operate in the real world. Conducting such studies



Figure 1: The Ommie Deployable System robot and companion tablet, developed for HRI supporting mental health.

requires replicable social robot systems that are autonomous, robust, adaptable by researchers, and customizable by users.

Simultaneously, a rapidly growing domain within in-the-wild HRI is the application of socially assistive robots for improving mental health and well-being [5]. This growth mirrors a staggering global rate of rising mental health challenges [14], and prior research shows that social robots can successfully support interventions for anxiety, depression, and wellness (e.g., [7, 13]). One robot expressly designed and developed for mental health applications is the Ommie robot [11]. We originally developed Ommie in 2021 for individuals with anxiety, using a human-centered design process and guidance from experts in psychology and neuroscience. The robot's core feature is a slowly expanding and contracting spherical body used to haptically guide deep breathing, which is known to calm the autonomic nervous system and reduce cortisol [8].

Ommie has been tested in multiple in-the-wild settings, including a university wellness center [12], children's hospital [3], and dementia care home [in peer review]. Results show the robot's ability to successfully guide deep breathing, provide calming effects, and reduce state anxiety. However, to study Ommie's impact on behavior change and skill development, the robot must be developed for longitudinal deployment in real-world settings.

We have accordingly created the *Ommie Deployable System (Ommie DS)*: a replicable, autonomous system for deploying a fleet of Ommie robots in user homes for weeks at a time (Fig. 1). Contextually, we imagine an individual with anxiety finding a calm, quiet space to place the robot in and taking time with it to practice deep breathing daily. Through practice, this skill becomes an



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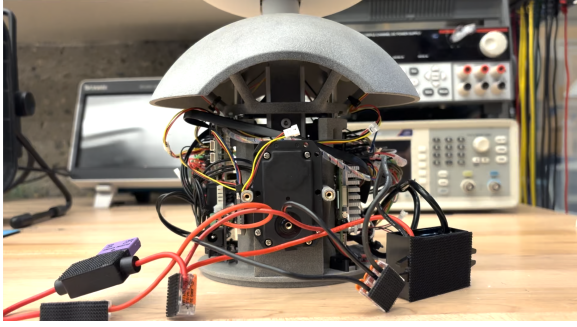


Figure 2: A view of the Ommie robot’s internal structure.

easily-reachable tool in times of high stress or anxiety. Ommie DS introduces new functional capabilities towards this aim beyond the initial prototype (Section 2, 3) and supports new robot interactions for deep breathing and user preferences (Section 4).

2 Hardware System

Ommie DS introduces major hardware improvements that address known failure points in prior versions and other deployable robots, as well as improving the end-user experience in real-world settings. To keep the system open and accessible to researchers, all electronics and mechanical components are off-the-shelf, custom enclosures are rapid-prototyped, and fixtures have been redesigned for easier assembly.

We implemented multiple mechanical robustness features, as well as materiality and component upgrades. For instance, the breathing motion with Ommie DS now uses mechanical rails and metal fixtures. The body’s shells are created with engineering-grade multi-jet fusion 3D printing to withstand falls, and its silicone covering is created with small-batch mold making. A robot “sweater” can be knit as a soft covering over the silicone. Two TFT displays continue to render expressive digital eyes in the robot’s head; however, to prevent light pollution in a dark bedroom, the screens (and other light-producing elements) are fitted with custom light-blocking gaskets. The audio of the robot has also been improved for stability with the use of a digital amplifier.

Ommie DS also contains an enhanced internal sensing suite, including a 4-way microphone array, an inertial measurement unit, temperature sensing, and ambient light sensing. Four capacitive touch sensors (on the robot’s belly, head, and the two shoulders) enable the user to haptically control the robot’s states – and for researchers to record hand placement during deep breathing. The robot also has a set of open data buses for easy addition of new sensors. A camera can be attached onto the robot’s head using a designed external adapter; however, we suggest caution due to the more private environments where users are likely to deep breathe.

The robot also now includes several practical components to support more seamless experiences in homes and remote technical debugging. For instance, the back of the robot features a physical power reset button and LED indicators for liveliness and error states. If needed, two small computer fans can be added inside a series of thermal vents for air flow. To enhance portability within the home (including between a table and lap), power is consolidated into a

single, removable 12 V barrel input with internal power management for all sub-components. For ease of access and modification, we have designed the external shells to be easily removable, with any constraining cabling attached via quick-disconnect fixtures.

Compute for Ommie DS is provided by a Raspberry Pi 5 (16 GB) onboard the robot and an optional wirelessly-connected computer box (e.g., a Minisforum MS-01, which allows for the addition of high-powered internal and external GPUs). A key part of the Ommie DS system is a wireless companion tablet (we use a Lenovo Tab One), which provides an additional UI for settings and controls.

3 Software System

The Ommie DS software ecosystem prioritizes future-proofing to support research longevity and wireless communications to improve end-user experiences and emulate real-world interactions. Our architecture is built on ROS2 Jazzy, Ubuntu 24.04, and Android 15 distributed across the three wirelessly networked devices: the robot (Raspberry Pi 5, Python 3.12), a local compute node (Linux Mini PC, Python 3.12), and a companion tablet (Android, Kotlin 2.0) (Figure 3). To ensure robust connectivity, the system utilizes the Zenoh protocol for middleware communication [1]. For remote fleet management, all devices connect via a private, global VPN, allowing researchers to monitor device status and deploy updates securely. The tablet interface features a custom Android Kotlin application that syncs data from the robot to expose the user’s current settings and interaction history, provide notifications from the robot, and enable user control of the robot’s states and setting of preferences. If wireless connectivity fails, the compute box can serve as a physical gateway between the tablet and robot.

Robot behaviors are orchestrated through a hierarchical Finite State Machine (HFSM) [4] designed for extensibility. At the lowest layer, individual ROS2 nodes manage hardware I/O, abstracting raw sensor data (e.g., capacitive touch, thermal readings) and actuator commands (e.g., motor control, screen animations). These are composed into mid-level behaviors, such as taking a deep breath or waking up from sleep. Completion and continuation states, such as *guided breathing* or *idling*, encapsulate specific behavior-calling logic and are managed by higher-level modes. These modes, such as *Do Not Disturb* or *Neutral*, enable the user to control when the robot is active or inactive. A high-level supervisory FSM governs the transitions between modes and states based on user inputs, scheduled wake and sleep windows, and sensor triggers. Overall, this modular design allows researchers to modify interaction flows without disrupting the underlying hardware control layers or recreating behavior logic from scratch.

To support safe and research-oriented home deployments, the system prioritizes data redundancy and health monitoring, as well as study-specific configuration and schedule tracking. User interactions are stored in a local database on the robot and backed up to the external compute box, while a core subset is streamed alongside system telemetry and logs to a secure cloud dashboard. An internal watchdog node monitors system-level failures, including overheating and low memory capacity, and notifies users via the robot and tablet to take any necessary action. Researchers can also configure their study’s schedule to specify deep breathing milestones and tablet notifications for completing study tasks.

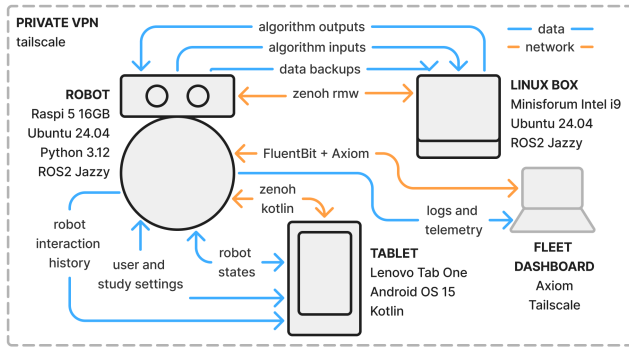


Figure 3: Ommie Deployable System networking and data passing.

4 Core Interactions

The Ommie Deployable System (DS) enables a set of interactions designed specifically for long-term, in-home mental health support. These interactions were co-developed with clinical psychologists to align with therapeutic practices, and are shaped by lessons from prior in-the-wild deployments of Ommie [3, 12].

Personalized Haptic Guided Breathing: At the center of Ommie’s interaction repertoire is its haptic breathing guidance. Our clinical partners emphasize that the most important component of deep breathing is practicing it regularly. We thus prioritize personalization, allowing users to find their preferred fit among a set of deep-breathing cadences (i.e., inhale, hold, and exhale lengths). Users can experiment on the tablet and change the robot’s default cadence at any time. They may also set “wake windows” around the time they want to practice deep breathing, during which the robot is allowed to wake up and attempt to engage with the user.

Breathing Skill Milestones: In long-term deployments, novelty effects pose a challenge [9], particularly for simple yet clinically powerful interactions such as deep breathing. To help sustain engagement over multiple weeks, we designed milestone-based interactions centered on the clinical concept of “joint celebration,” in which the therapist and patient socially share a moment of celebration to reinforce an accomplishment. When users reach meaningful points in their breathing practice (e.g., completing their first session, returning after an absence, or achieving a streak), the robot performs special celebratory behaviors. The tablet also surfaces a notification and logs the milestone in the user’s breathing history. Rather than gamifying mental health, the intent is to acknowledge incremental progress in a relational manner that encourages continued practice.

Preferences for Robot Placement and Proactivity: Since deep breathing is often practiced in quiet, private settings, Ommie DS is intentionally designed to be flexible with respect to context and user control. Unlike many in-home robot systems that rely on fixed docking locations or wired connections, our architecture allows the robot, companion tablet, and external compute node to be placed anywhere in the home. Users can move the robot to their bedside, couch, or floor for maximal comfort and can relocate noisier compute hardware (e.g., with a GPU fan) elsewhere. Prior studies with home-deployed robots have also indicated that users can find robot proactivity disruptive [2, 6]. With Ommie DS, users also maintain

granular control over when the robot is active via configurable *Do Not Disturb* and *Wake* mode time windows, which can be scheduled or toggled at any time through the tablet.

5 Demonstration

Our demonstration will include a fully functioning Ommie DS with which users can experience guided-breathing behaviors and the companion tablet. We will also present a developer-focused robot that showcases its internal mechanisms, sensing, and the design choices that enabled the recent creation of a fleet of ten Ommie DS units. We additionally aim to use this demonstration as a forum for gathering community input towards enabling outside replication.

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